

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I A. Lakshmi certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Lakshmi*

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Answers

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice

Introduction:

Virtualization is the foundation of cloud computing. A hypervisor is software that creates and manages multiple virtual machines (VMs) on a single physical server. Hypervisors are mainly divided into Type-1 (Bare-metal) and Type-2 (Hosted) hypervisors. For mission-critical applications such as banking systems, hospital databases, airline reservation systems, or military software, the choice of hypervisor directly affects system performance, reliability, security, and maintenance.

Comparison:

Factors	Type 1	Type 2
Installation	Runs directly on physical hardware	Runs on top of an operating sys
Performance	Very high	Lower due to OS overhead
Security	Storage security	More vulnerable
Resource Usage	efficient	Consumes additional resources
Reliability	Very stable	Depends on host operating system
Management	Enterprise management tools	Mainly desktop usag
Example	VMware Workstation, VirtualBox	VMware Workstation, VirtualBox

Performance Analysis

Type-1 hypervisors communicate directly with CPU, memory, storage, and network devices without an intermediary operating system.

Advantages include:

- Lower CPU overhead
- Faster memory access
- Better I/O throughput
- Faster VM scheduling
- Higher scalability

For example,

Suppose a cloud server hosts 200 banking transaction servers.

A Type-1 hypervisor can efficiently allocate CPU time among VMs, whereas a Type-2 hypervisor wastes resources because every VM request first passes through the host operating system.

Result:

- Higher throughput
- Lower latency
- Better response time

Security Analysis

Type 1: Security Analysis

Security is the biggest concern for mission-critical systems.

Type-1 Hypervisor

Because there is no host operating system,

- Smaller attack surface
- Fewer vulnerabilities
- Better isolation
- Reduced malware exposure

Even if one VM is compromised, other VMs remain isolated

Type-2 Hypervisor

Since it depends on a host operating system,

Attack path becomes:

Hardware, Host Operating System ,Hypervisor and Virtual Machines

If malware infects the host OS, every VM becomes vulnerable.

Therefore,

Type-2 introduces additional security risks.

Manageability Analysis

Modern Type-1 hypervisors support

- Live migration
- High Availability (HA)
- Fault tolerance
- VM snapshots

- Centralized management
- Automated load balancing
- Disaster recovery

These features are essential for enterprise cloud environments.

Type-2 hypervisors provide only basic VM management and are mainly used for development or testing.

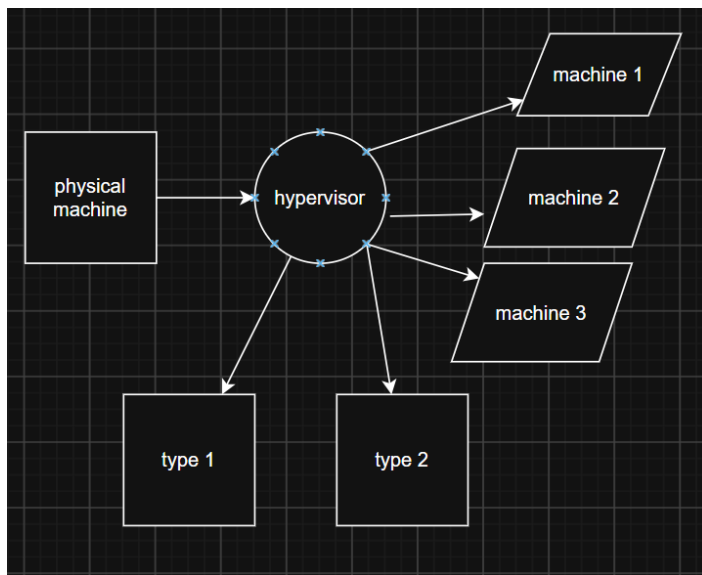
Final Justification

For mission-critical cloud applications,

Type-1 Hypervisor is the better choice because

- Highest performance
- Minimal overhead
- Better scalability
- Better VM isolation
- Enterprise-level security
- High availability support
- Easier centralized management

Related Diagram



2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Virtualization:

The operation where it converts the one physical server into multiple servers through the help of the hypervisor.

Problem Analysis:

A cloud provider wants to host

- Finance applications
- Healthcare systems
- Student portals

on the same physical hardware.

These applications have different security requirements.

Finance requires:

- Confidentiality
- Integrity
- Regulatory compliance

Healthcare requires:

- Patient privacy
- Medical data protection

Student portals require:

- Multi-user access
- Moderate security

How Virtualization Provides Isolation

1. CPU Isolation

Each VM receives dedicated virtual CPUs.

The hypervisor schedules CPU time independently.

A finance VM cannot directly execute processes inside the healthcare VM.

2. Memory Isolation

Each VM receives its own virtual memory.

Memory pages are protected using hardware-assisted virtualization.

Example

Finance VM

Memory Range:

1000–3000

Healthcare VM

Memory Range:

5000–9000

The finance VM cannot read healthcare memory.

3. Storage Isolation

Each VM owns separate virtual disks.

Access permissions prevent unauthorized reading.

Even though physical disks are shared,
logical storage remains isolated.

4. Network Isolation

Virtual switches

Virtual LANs (VLANs)

Software Defined Networking

ensure traffic separation.

Finance traffic never mixes with healthcare traffic.

5. Access Control

Different administrators receive different privileges.

Healthcare administrators cannot modify finance VMs.

Two Major Attack Surfaces

Attack Surface 1: Hypervisor Attacks

If attackers exploit a hypervisor vulnerability,
they may gain access to multiple VMs.

Possible consequences

- Data theft
- VM takeover
- Service disruption

Attack Surface 2: VM Escape

VM escape occurs when malicious software breaks out of a virtual machine and gains control over the host or neighboring VMs.

This is considered one of the most dangerous virtualization attacks.

Example

Student portal compromised, Attacker escapes VM, Accesses finance server and Financial information stolen

Conclusion

Virtualization provides strong isolation through

- CPU separation
- Memory isolation
- Storage isolation
- Network segmentation
- Access control

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Given Snapshot

CPU Usage = 92%

Memory Usage = 88%

Storage I/O Latency = 35 ms

Average VM Boot Time = 170 seconds

Step 1: CPU Analysis

CPU utilization above 90% indicates that processors are almost fully occupied.

Effects:

- Slow application response
- Increased VM scheduling delays
- High context switching
- Performance degradation

Likely cause:

Too many VMs competing for CPU resources.

Corrective action:

- Migrate some VMs to another host
- Add more CPU cores
- Limit CPU-intensive workloads
- Enable dynamic resource scheduling

Step 2: Memory Analysis

88% memory utilization is very high.

When physical memory is nearly exhausted,
the hypervisor may begin memory reclamation techniques such as:

- Ballooning
- Compression
- Swapping

Swapping significantly slows VM performance because data is moved between RAM and disk.

Corrective action:

- Increase RAM capacity
- Reduce VM density
- Optimize memory allocation

Step 3: Storage I/O Latency Analysis

Latency = 35 ms

For enterprise applications, storage latency is typically expected to remain below 10–20 ms. A value of 35 ms indicates a storage bottleneck.

Possible causes:

- Slow HDD storage
- Too many simultaneous disk requests
- Storage controller congestion
- Heavy database operations

Effects:

- Slow database queries
- Delayed file access
- Increased VM response time

Corrective action:

- Upgrade to SSD or NVMe storage
- Implement storage tiering
- Balance storage workloads
- Improve storage controller performance

Step 4: VM Boot Time Analysis

Boot Time = 170 seconds

Normal enterprise VM boot times are generally much lower.

Possible causes:

- CPU overload

- Memory shortage
- High storage latency
- Slow operating system initialization

Overall Bottleneck Analysis

The system is suffering from resource contention, where multiple VMs compete for limited CPU, memory, and storage resources.

The high CPU and memory utilization reduce processing capacity, while excessive storage latency delays data access. These factors together lead to unusually long VM boot times and poor application performance.

Final Recommendations

- Add additional physical hosts to distribute workloads.
- Increase CPU cores and RAM.
- Migrate heavily loaded VMs to less utilized hosts.
- Replace HDDs with SSDs or NVMe drives.
- Enable automated load balancing and resource monitoring.
- Continuously monitor resource utilization to prevent future bottlenecks.

4. How Software Defined Datacenter (SDDC) Improves Agility

Introduction

A traditional datacenter requires administrators to manually configure servers, storage, and networking. This process is time-consuming, error-prone, and difficult to scale.

A Software Defined Datacenter (SDDC) virtualizes and manages compute, storage, and networking entirely through software. Resources can be provisioned, scaled, and managed automatically using centralized policies.

Compute Virtualization

In a traditional datacenter:

- Applications are tied to physical servers.
- Adding new servers requires manual installation and configuration.

In an SDDC:

- Physical servers are abstracted into virtual machines.
- New VMs can be deployed in minutes.
- Workloads can be migrated without downtime.

Benefit: Faster deployment and improved resource utilization.

Storage Virtualization

Traditional storage:

- Fixed disks with manual allocation.
- Difficult to expand and manage.

SDDC:

- Combines multiple storage devices into a unified storage pool.
- Capacity can be allocated dynamically based on application needs.
- Supports snapshots, replication, and automated backups.

Benefit: Flexible storage management with better resilience.

Network Virtualization

Traditional networks:

- Routers and switches require manual configuration.
- Network changes are slow.

SDDC:

- Uses Software Defined Networking (SDN).
- Networks, firewalls, and load balancers are created through software.
- Security policies can follow workloads automatically.

Benefit: Rapid network provisioning and consistent security.

Overall Agility Improvements

Compared to traditional Datacenters , SDDCs provide:

- Rapid provisioning of infrastructure
- Automated resource scaling
- Centralized management
- Better disaster recovery
- Faster deployment of new applications
- Higher operational efficiency
- Reduced manual errors

These capabilities allow organizations to respond quickly to changing business needs while maintaining high availability and efficient resource utilization.

5.A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Multi-Cloud Federation Identity Mismatch

Problem Analysis

A company uses multiple cloud providers for different services. During integration, users experience authentication failures because each provider maintains separate identity systems, user IDs, authentication methods, and access policies.

As a result:

- Users must maintain multiple accounts.
- Authentication becomes inconsistent.
- Access management becomes complex.
- Security risks increase due to duplicate credentials and misconfigured permissions.

Why Identity Mismatches Occur

Different cloud providers may:

- Store user identities independently.
- Use different authentication protocols.
- Maintain separate authorization policies.
- Support different user attribute formats.

This leads to synchronization issues and inconsistent access control across environments.

Secure Identity Federation Design

A secure federation model introduces a **central Identity Provider (IdP)** that authenticates users once and establishes trust with all participating cloud providers.

Authentication Flow

1. The user signs in to the Identity Provider.
2. The Identity Provider verifies the user's credentials.
3. A trusted authentication token is issued.
4. Cloud providers validate the token.
5. The user gains authorized access without creating separate accounts.

This enables **Single Sign-On (SSO)**, allowing users to authenticate once and securely access multiple cloud environments.

Privacy Protection Measures

To safeguard user privacy:

- Share only the minimum identity attributes required by each service.
- Encrypt authentication tokens during transmission.
- Enforce Multi-Factor Authentication (MFA) for sensitive workloads.
- Apply Role-Based Access Control (RBAC) to ensure users receive only the permissions necessary for their roles.
- Maintain audit logs to track authentication and access events.
- Use short-lived tokens and automatic session expiration to reduce the impact of credential compromise.

Benefits of Identity Federation

- Single Sign-On improves user experience.
- Centralized identity management simplifies administration.
- Consistent security policies across cloud providers.
- Reduced password duplication and credential sprawl.
- Stronger privacy through controlled sharing of identity information.